



Figure 1: Web content extraction – Python libraries

```
proxies={'http':'10.10.80.11:3128','https':'10.10.80.11:3128'}
url = "http://opensourceforu.com"
res = requests.get(url, proxies = proxies, timeout=5.0 )
html = res.text
# extract the text from the html version
text = html2text.html2text(html)
print(text)
```

Lassie

The Lassie library enables the retrieval of component information from Web pages. For example, the key words, images, videos and description can be fetched with the Lassie library. The installation of Lassie can be carried out with the following commands:

```
$ pip install lassie
```

...Or,

```
$ easy_install lassie
```

The information from the Web page is gathered with the *lassie.fetch()* function as illustrated below:

```
import lassie
import pprint
url="http://opensourceforu.com/2016/04/ubuntu-16-04-lts-xenial-xerus-official/"
pprint.pprint (lassie.fetch(url))
```

The execution of the above script results in the following output:

```
{'description': 'u'Ubuntu 16.04 LTS aka "Xenial Xerus" comes with new features like snap package and LXD container to deliver you an upgraded experience. Being as an LTS platform, the new version additionally comes with five-year support.',
```

```
'images': [{'src': 'u'http://opensourceforu.com/wp-content/uploads/2016/04/Ubuntu-16-04-lts-150x150.jpg',
            'type': 'u'og:image'},
          {'src': 'u'http://www.opensourceforu.com/favicon1.ico',
```

```
'type': 'u'favicon'}],
'keywords': [u'Ubuntu',
             u'Ubuntu 16.04 LTS',
             u'Ubuntu 16.04',
             u'Ubuntu Xenial Xerus',
             u'Xenial Xerus',
             u'Canonical'],
'locale': 'u'en_US',
'title': 'u'Ubuntu 16.04 LTS "Xenial Xerus" goes official - Open Source For You',
'url': 'u'http://opensourceforu.com/2016/04/ubuntu-16-04-lts-xenial-xerus-official/',
'videos': []]
```

Lassie facilitates the fetching of all the images from the Web page with the *all_images = True* option. The following code will provide information regarding all the images present in the page:

```
pprint.pprint (lassie.fetch(url, all_images=True))
```

Newspaper

The Newspaper library, developed and maintained by Lucas Ou-Yang, is specially designed for extracting information from the websites of newspapers and magazines. The objective of this library is to extract and curate the articles from the newspapers and similar websites. To install Newspaper, use the following commands.

For lxml dependencies, type:

```
$ sudo apt-get install libxml2-dev libxslt-dev
```

PIL is for recognising images.

```
$ sudo apt-get install libjpeg-dev zlib1g-dev libpng12-dev
```

For Newspaper library installation, type:

```
$ pip3 install newspaper3k
```

The NLP corpora will be downloaded:

```
$ curl https://raw.githubusercontent.com/codelucas/newspaper/master/download_corpora.py | python3
```

The Newspaper library is used to gather information associated with articles. This includes author names, publication dates, major images in the article, videos present in the article, key words describing the article and the summary of the article. A sample program with an article from the *Open Source For You* magazine is given below: